Hitchin Harmonic Spinors Notes

1 Outline

1. **Preliminaries**
   (a) Clifford algebra
   (b) spin groups and representation from last time
   (c) lifting connection to spin
   (d) spin c
2. theorem statement and proof

Let V be an even dimensional spin vector bundle. If we equip V with a connection ∇, we get a corresponding connection on the principal-$\mathrm{SO}(2n)$ bundle of frames of V, denoted P_{SO}. This lifts to a connection on P_{Spin}, the principal-spin$(2n)$ bundle on V, by the covering map.

One can show that if the connection 1-form is given locally by ω_{ij}, then relative to this lifting, the lifted matrix is given by

$$\frac{1}{4}\sum_{i,j}\omega_{ij}e_ie_j$$